Progress Report: Academic Assistant via DPO, MCQA, Quantization, RAG

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1 Introduction

We implemented all four components—MCQA, DPO, quantization, and RAG—and conducted initial experiments: fine-tuning MCQA with/without Chain-of-Thought (CoT), training DPO on filtered dataset variants, applying quantization, and training RAG on a curated corpus.

2 Data

2.1 Prompting strategies for M1 Dataset

Table 1 outlines each member's A/B prompt design, including structure, persona, and contrast method.

Name	Prompting Strategy
Clémentine	Created four templates for MCQs and open-ended questions, generating both high- and lower-quality responses in the same chat. A emphasized step-by-step reasoning; B was plausible but less structured or incomplete. Prompts were adjusted for edge cases.
Shuli	Used separate chats to avoid memory effects. A applied chain-of-thought reasoning; B prioritized brevity with length limits. Reran prompts if responses were unbalanced in quality.
Kyuhee	Refined A/B prompts over four iterations with a custom gold-answer checker. A used a STEM assistant persona with formal reasoning, later shortened to reduce verbosity. B used a tutor persona focused on clarity and brevity.
Stergios	Reused one chat per course, reinitializing for A and B. A used a teaching assistant persona with step-by-step reasoning; B was framed as a friend giving concise help. Prompts were adapted per question type with full options.

Table 1: Prompting Strategies for Milestone 1

2.2 Reward Model

We evaluated our DPO models on the MNLP-DPO evaluation set[1]. To assess data quality impact, we split the **M1** dataset[2] into four filtered variants based on consistency and ambiguity. Following the original setup, we included UltraFeedback (Cui et al., 2023)[3] (61.1K) and added Code-Preference-Pairs (Vezora, 2024) (54K) for topical alignment.

Dataset	Subset	Size	Description
M1 Dataset	v1	24.2K	Overall preference only (A/B only)
	v2	19.7K	Overall pref. (A/B only) + other pref. consistent (AB allowed)
	v3	23.2K	Overall pref. (A/B only) + correctness pref. match (AB allowed)
	v4	3K	All criteria consistent (A/B only)

Table 2: Datasets used for DPO training

2.3 MCQA and Quantization Models

We evaluated our MCQA and quantized models on the MNLP-STEM-mcqa set[4]. The MCQA model was fine-tuned on a mix of public datasets (OpenBookQA (Mihaylov et al., 2018), SciQ (Johannes Welbl, 2017), QASC (Khot et al., 2020)) and filtered **M1** MCQs, all standardized and preprocessed. For quantization, 512 samples from the same sources were used for calibration. See Appendix A for details.

2.4 RAG Model

The RAG model was trained on a corpus combining the **M1** dataset, Natural Questions (Kwiatkowski et al., 2019), KILT (Petroni et al., 2021), and papers from arXiv, standardized as QA pairs or document chunks. See Appendix B for details.

3 Training

3.1 Reward Model

Reward models were trained using `Qwen3-0.6B-Base` and `trl`'s `DPOTrainer` for 3 epochs (batch size 2, grad accumulation 4, FP16). Inputs were truncated to 128 tokens

[1] https://huggingface.co/datasets/zechen-nlp/MNLP_dpo_evals

[2] Student-contributed preference data collected during Milestone 1.

[3] Binarized version: https://huggingface.co/datasets/HuggingFaceH4/ultrafeedback_binarized

[4] https://huggingface.co/datasets/zechen-nlp/MNLP_STEM_mcqa_evals

(prompts) and 512 (full sequences). We compared a baseline (LR 1×10^{-5}, no clipping) and a tuned setup (LR 5×10^{-6}, `max_grad_norm` = 1.0). Training dynamics are in Appendix C.

3.2 MCQA and Quantized Model

MCQA models were fine-tuned using `trl`'s `SFTTrainer`, with and without CoT prompts[5], using batch size 1, grad accumulation 8, LR 5×10^{-5}, 512-token limit, FP16, and 1 epoch. The CoT model was quantized via 8-bit PTQ using `llm-compressor` (Neural Magic, Inc., 2025) with SmoothQuant+GPTQ (smoothing 0.8).

3.3 RAG Model

The RAG retriever uses `all-MiniLM-L6-v2`[6], trained with `ContrastiveTensionLoss` for unsupervised segment learning. We fine-tuned `Qwen3-0.6B-Base` via `SFTTrainer` (batch size 2, grad accumulation 4, 1 epoch, LR 2e-4, AdamW).

4 Preliminary Results

4.1 Reward Model

Figure 1 shows reward accuracy on the MNLP-DPO benchmark for models trained on different dataset variants.[7] The best result came from the `v2` split with tuned hyperparameters (87.5%), highlighting the impact of data quality and careful tuning given its smaller size. `UltraFeedback` (87.3%) and `Code-Preference-Pairs` (86.9%) also performed well. Other variants showed modest declines, while `v4` dropped sharply to 71.2%, likely due to reduced training data from strict filtering.

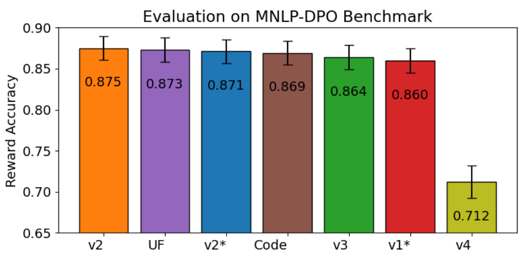

Figure 1: Reward accuracy on MNLP-DPO. All models use the tuned setup unless marked * (original). UF = UltraFeedback, Code = Code-Preference-Pairs.

[5] CoT applied only to public data, as M1 lacks rationales.

[6] https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2

[7] Results on demo datasets (`zechen-nlp/MNLP_dpo_demo`) are provided in Appendix D.

4.2 MCQA and Quantization Models

Figure 2 shows MCQA accuracy across setups. The base model scored 31.4% without fine-tuning. Fine-tuning raised it to 35.97% with CoT and 40.78% without, suggesting that added rationales may introduce noise.[8] The quantized CoT model retained similar performance (36.1%)[9] while significantly improving efficiency—reducing GPU memory usage from 2.3 GB to 1.2 GB and model size from 22.8 GB to 0.73 GB—at the cost of roughly 5× slower inference.

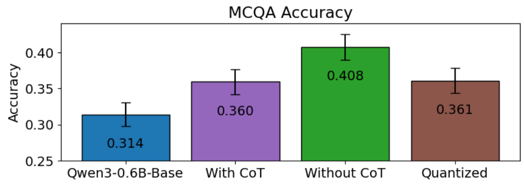

Figure 2: MCQA performance comparison.

4.3 RAG Model

The RAG model is currently incompatible with the eval suite (see `ED #505`); we plan to address this.

5 Future directions

5.1 Reward Model

We plan to combine datasets to test complementarity and automate filtering to improve quality. Additionally, we will expand evaluation with diverse test sets and refine pre-DPO initialization and hyperparameters for stronger alignment.

5.2 MCQA and Quantized Models

We will try filtering for single-answer questions and enhance CoT-based training by incorporating varied reasoning styles and generating rationales for **M1** MCQs. For quantization, we will explore alternative calibration and smoothing strategies, and compare PTQ with Quantization-Aware Training.

5.3 RAG Model

For RAG, we plan to scale training with additional datasets, refine document selection, and retrain the Qwen backbone to improve retrieval. Since the current model is incompatible with the evaluation suite, we will prioritize building a compliant pipeline to enable proper assessment.

[8] On the 50-sample demo set (`zechen-nlp/MNLP_STEM_mcqa_demo`), the trend was consistent: 0.28 ± .0641 (base), 0.30 ± 0.0655 (with CoT), and 0.38 ± 0.0693 (without CoT).

[9] Demo set: 0.300 ± 0.0655.

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A MCQA and Calibration Dataset Details

Dataset	Size	Description
Student-contributed	789	Filtered MCQA only; no CoT rationales
OpenBookQA (Mihaylov et al., 2018)	4.96K	Elementary science questions with one correct answer
SciQ (Johannes Welbl, 2017)	11.7K	Crowd-sourced science questions with distractors
QASC (Khot et al., 2020)	8.13K	Multi-hop science questions derived from QA pairs

Table 3: MCQA training datasets used for model fine-tuning and quantization calibration. Statistics reflect 90% of the original data, with 10% held out for validation.

B RAG Dataset Overview

Dataset	Collected	Used for Training	Description
Internet-sourced (arXiv)	4,765	4,765	Research paper chunks
Student-contributed	1,264	1,264	One answer per question
Natural Questions	300K	tested with up to 2,000 (0 for m2 submission)	Filtered Q&A pairs
KILT Dataset	9M	tested with up to 5,000 (0 for m2 submission)	Wikipedia passages

Table 4: Collected and used datasets for RAG model training.

C DPO Training Curves

We include training metrics to illustrate optimization trends under different dataset and configuration settings.

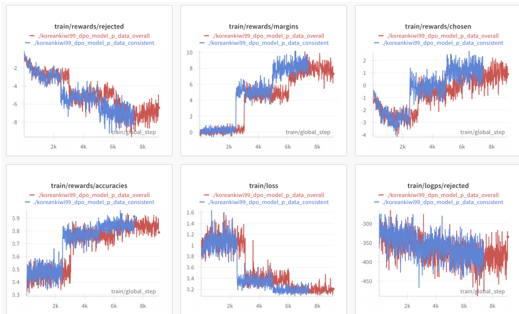

Figure 3: Training dynamics on v1 vs. v2 (original configuration). v2 shows better margin separation and reward accuracy.

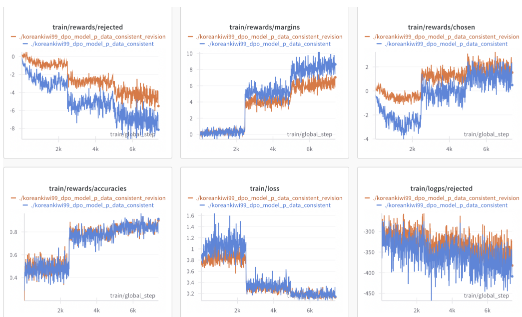

Figure 4: Training dynamics on v2 using tuned hyperparameters (LR $1 \times 10^{-5} \rightarrow 5 \times 10^{-6}$ and gradient clipping). Improvements are seen in stability, margin growth, and reward trends.

Figure 5: Training dynamics on `v2`, `UltraFeedback`, and `Code-Preference-Pairs`. Notably, the Code dataset shows highly atypical reward signals, while UltraFeedback tracks closely with student data.

D DPO Results on Demo Dataset

On the 50-sample demo set, `v4` achieved the highest reward accuracy (0.68), followed by `UltraFeedback` (0.62) and `Code-Preference-Pairs` (0.58). In contrast, `v2`—despite being the best overall on the full evaluation set (87.5%)—scored lower here (0.56). This reversal suggests a distribution shift between the demo and full sets, with `v4` possibly better aligned to the demo data. Due to the small sample size, results should be interpreted cautiously and viewed as complementary to the main benchmark.

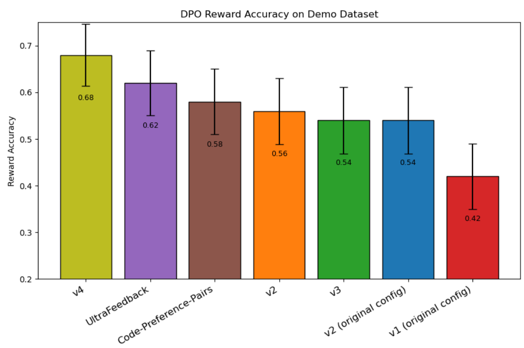

Figure 6: Reward accuracy of DPO models on the MNLP-DPO demo dataset.